

Homework (Habanero)

- Q1.** What is the axis of symmetry for the parabola $y = x^2 + 4x + 4$?
- (A) $x = -2$
 (B) $x = 0$
 (C) $x = 2$
 (D) $x = 4$
 (E) $x = -4$
- Q2.** A new medicine dosage is modeled by the quadratic $y = -0.5x^2 + 3x + 2$. What is the maximum dosage?
- (A) $y = 4.5$
 (B) $y = 5.5$
 (C) $y = 6.5$
 (D) $y = 7.5$
 (E) $y = 8.5$
- Q3.** The vertex of the parabola $y = x^2 - 6x + 8$ is at
- (A) $(-3, 1)$
 (B) $(-2, 4)$
 (C) $(-1, 7)$
 (D) $(3, -1)$
 (E) $(2, 0)$
- Q4.** The quadratic $y = 2x^2 - 4x - 6$ has y -intercept at
- (A) $(0, -2)$
 (B) $(0, -4)$
 (C) $(0, -6)$
 (D) $(0, 2)$
 (E) $(0, 6)$
- Q5.** A population of cells grows according to the quadratic $y = 3x^2 + 2x + 1$. What is the rate of change of the population at $x = 2$?
- (A) $y' = 12$
 (B) $y' = 14$
 (C) $y' = 16$
 (D) $y' = 18$
 (E) $y' = 20$
- Q6.** What is the equation of the axis of symmetry for the parabola $y = x^2 + 2x - 6$?
- (A) $x = -1$
 (B) $x = 0$
 (C) $x = 1$
 (D) $x = 2$
 (E) $x = 3$
- Q7.** A medical researcher models the spread of a disease using the quadratic $y = -x^2 + 4x + 2$. What is the minimum value of y ?
- (A) $y = -2$
 (B) $y = 0$
 (C) $y = 2$
 (D) $y = 4$
 (E) $y = 6$
- Q8.** The parabola $y = x^2 - 4x + 4$ opens
- (A) upward
 (B) downward
 (C) to the left
 (D) to the right
 (E) not at all

Open Ended Questions (FRQ)

- Q9.** Prove that the vertex of the parabola $y = ax^2 + bx + c$ is at $(-\frac{b}{2a}, f(-\frac{b}{2a}))$. Show all work and explain your reasoning.
- Q10.** A new medicine is modeled by the quadratic $y = -0.2x^2 + 2x + 1$. Find the maximum dosage and the time at which it occurs. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use graphing calculators to visualize the parabolas.
- Tip 2: Remind students to show all work and explain their reasoning for free response questions.
- Tip 3: Emphasize the importance of axis of symmetry and vertex in graphing quadratics.